

**STATEMENT OF WORK (SOW)
FOR
An EO/IR Scene Generation System
RESEARCH, DEVELOPMENT, TEST & EVALUATION (RDT&E) SUPPORT**

1.0 PROJECT OVERVIEW

1.1 Introduction

Through the Central Test & Evaluation Investment Program (CTEIP) Electro-Optical Infrared (EO/IR) Direct Injection Project, the Naval Air Warfare Center – Integrated Battlespace Simulation and Test Department (NAWC-IBST) intends to develop the capability to directly inject EO/IR imagery into systems under test (SUT) for fixed wing and rotary winged platforms. The F-35 is intended to be the first of several platforms introduced to this capability under this scope of work. The injection capability will enable the F-35 aircraft and rotary wing platforms to test in a variety of mission-representative scenarios comprehensively evaluating the effectiveness of the ICP with the EO/IR sensor suites across various platforms and augmenting flight test with repeatable, complex EO/IR scenarios. This work will require access to, and handling of, controlled unclassified information.

The Navy is the lead Service coordinating the EOIRDI project, which is incrementally funded by the Office of the Secretary of Defense (OSD) Test Resource Management Center (TRMC), the sponsor of the EOIRDI project. The Air Force and Army are the deputy Services supporting the EOIRDI project. NAWC-IBST is responsible for the technical management of the EOIRDI project and will provide guidance and oversight to coordinate the system design, development, integration, test and validation delivery activities of the EOIRDI integrated project team (IPT) for the EOIRDI effort.

1.2 Purpose

This Statement of Work (SOW) defines the breadth and scope of the effort to analyze and document the EO/IR scene generation (SG) systems (SGS) build and deployment to support the developmental testing and functional system integration of the overall EOIRDI system, as well as the continued development of existing model databases for the enhancement of scene fidelity and scenario development methodology, as defined in the EOIRDI Performance Specifications Document. The SGS capability must demonstrate both Blocks A and B capabilities upon contract award, with no additional hardware or software development required to satisfy SUTs-supported requirements, as identified in the EOIRDI Performance Specifications Document. EOIRDI development, testing, and integration should not require any modification to the core SGS capability and is limited to model databases and scenario development methodology, as defined in the EOIRDI Performance Specifications Document. The resulting products delivered shall be a collaboration between the Contractor and the GOVT IPT of NAVAIR IBST at NAS Patuxent River, MD.

The effort required to develop and deliver other EOIRDI components and capabilities external to the SGS (scenario executive, external stimulation controller, or SUT sensor emulation) will be performed separately and is not included within the scope of this SOW.

2.0 DOCUMENTS

The following list of documents apply to all tasking under this contract. Additional and/or specific requirements may be detailed in each Technical Direction Letter (TDL).

DoDM 5200.01	DoD Information Security Program
MIL-STD-961	DoD Standard Defense and Program-Unique Specifications Format and Content
SPEC	Electro-Optical Infrared (EO/IR) Direct Injection (EOIRDI) Performance Specifications Document, Test Capability Requirements for the Scene Generation System

3.0 REQUIREMENTS

3.1 Program and Engineering Management

The Contractor shall provide the program management to manage the non-recurring engineering and design activities required to deliver the scope of work defined in this SOW and SPEC. The contractor shall provide the infrastructure to support open dialog with the gov't team to manage schedule and scope.

The contractor shall provide initial integrated master schedule (IMS) and updates to meet program milestones (MS). The contractor shall provide monthly status reports (MSR) for project execution and expenditure tracking. The contractor shall provide plans to execute development and capability demonstrations as detailed under the Scope of this SOW and accompanying Specifications Document. The contractor shall coordinate, support, and document success criteria of each MS event. The Contractor shall support Risk Management Framework (RMF) and Requirements Traceability IAW DoD standards.

The contractor shall address the development for the Scene Generation System for operation for use within the EO/IR direct inject capability being developed for the EO/IR direct inject project. The Scene Generation System configuration and requirements are intended for use with multiple fixed and rotary winged Systems under Test (SUT) of Air Force, Army, and Naval platforms.

NAWC-IBST is responsible for the technical management of the EOIRDI project. NAWC-IBST will provide guidance and oversight to coordinate the system design, development, integration, test and validation delivery activities of the EOIRDI integrated project team (IPT) for the EORIDI effort.

3.2 Project Scope

The scope of this effort accommodates the delivery and site integration of two (2) site-specific Scene Generation Systems for final deployment at two (2) GOVT sites. The two (2) of these SGS deliverables are to be deployed during the Block A program phase for site integration at NAS Patuxent River NAS and either Eglin or Edwards Air Force Base. Execution and development must efficiently combine and/or segregate efforts for F-35 (Block A) and RW (Block B) platforms supported, as appropriate, to minimize duplication of efforts and maximize execution and productivity.

The EOIRDI project requires the incremental delivery of assets to support their use as GFE test assets during platform interface development, with continued SW development and incremental upgrades to SGS to accommodate deployment schedule, projected milestones, and full final capability of all SG assets included under this scope. SG production schedule must prioritize the accommodation of the platform interface development schedules in order to successfully arrive at site acceptance IAW EOIRDI system development. SGSs developed under this effort will be initially accepted by the GOVT for GFE deployment as developmental test assets for platform interface development, and finally accepted by the GOVT to meet EOIRDI system requirements upon site integration. Continuing SW development and upgrades shall be performed in increments as detailed in the EOIRDI Performance Specifications document and this SOW, to meet the capability requirements of the final deliverable.

Integrated system interim capability demos are planned throughout project development.

Note the following assumptions:

- CLIN 1 will occur first
- CLINs 2, 4, and 6 are software upgrades used by CLIN's 1, and 3
- CLINs 3, 4, 5, and 6 are optional CLINs
- CLIN 7 & 8 are for spares
- Systems delivered under CLIN's 1, and 3 are for different customers; not all CLIN's may be awarded
- CLIN 7 & 8 are reserved

CLIN 1 – Block A – 6-Channel PAX System, Support Material, Long Lead Material Procurement, Baseline/Tier 0 SW & Tier 1 SW

1. System Design and Configuration. IAW Appendix A:

- a. Perform SRR of Scene Generation System operational specifications and Tier 1 modeling development, to include engineering pathway to Tier 2 modeling requirements. Requirements shall be traceable to the Performance Specifications Document and the use case scenarios supporting overall system capability. (SRR A002, RTVM A008 CDRLs)
- b. Perform PDR of SG System Design Configuration with Tier 1 development, to include Baseline SKIP capability, EOIRDI connectivity requirements, and updated RVTM

- with test planning for interim modeling capability achievements. (PDR A003, RVTM A008 CDRLs)
- c. Perform CDR of final SG System configuration with Tier 1 development, and updated accompanying test plans for interim modeling capability achievements to include preliminary considerations toward Tier 2 development. (CDR A004, Test Plan & Procedures A014 CDRLs)
 - d. Demonstration of hardware and software system capability compliance IAW the Scene Generator Performance Requirements defined in the EOIRDI Performance Specifications Document (FAT Report A015 CDRL, SAT Report A016 CDRL)
2. Build of SG Systems
- a. Capability status check of Baseline SKIP capability and EOIRDI integration compliance prior to procurement.
 - b. Build of SG systems to include two (2) Control and Modeling PC and 8 image generation channels, configured to support both Blocks A and B operational capability
 - i. Configuration for Block A operational capability
 - 1. One (1) Control and Modeling PC
 - 2. Six (6) image generation channels
 - ii. CLIN 1 Spares
 - 1. One (1) Control and Modeling PC
 - 2. Two (2) image generation channels
 - c. FAT of the scene generators and control and modeling PCs identified in item 2.c. The SG system Facility Acceptance Testing (FAT) procedure shall be performed on each completed asset before transfer to EOIRDI project sites. The purpose is to verify full functionality of SKIP using standard models and scenarios at 60Hz, 500Hz and 1000Hz, and frame size up to 2048x2048, in standalone and external control mode, with Use Case developed scenarios and signature validation, IAW SGS Performance requirements defined by the EOIRDI Performance Specifications Document.
 - d. Transfer of 1 Control and Modeling PC and two Image Generation Channels to NG-Amherst for developmental work
 - e. Transfer of 1 Control and Modeling PC and 6 Image Generation Channel to Lockheed Martin Corporation at Fort Worth for developmental work
 - f. Capability status check of Scene Generation System with 6 channels with Use Case developed scenarios, and signature validation at EOIRDI Development site
3. Tier 1 Model Application Development for baseline system and future growth
- a. Design and develop Tier 1 Model Application for baseline system IAW EOIRDI Performance Specifications Document
 - b. Perform interim Tier 1 capability demonstration event.
 - c. Perform Tier 1 capability demonstration and FAT IAW Tier 1 requirements defined in the EOIRDI Performance Specification Doc and the RTVM, prior to integration with EOIRDI system. The FAT shall be performed using the Tier 1 model capability to demonstrate predefined EOIRDI SUT uses case scenarios with threat fly-out, and threat infrared signature models to confirm scene background, clutter, and atmospheric characteristics, as well as to confirm compatibility with the Baseline SKIP SGS modeling capabilities.
 - d. Perform Tier 1 model updates and integration for SG systems and integration as required during model application developmental cycle

- e. Perform Tier 1 model demonstration during EOIRDI system FAT and IOC.
 - f. Base 6-channel system will be delivered to Lockheed Martin with baseline SKIP functionality. Model application development capabilities may not be developed in time to support Block A capability development at Lockheed Martin Corporation, in which case, the EOIRDI system deliverable shall be updated prior to PAX SAT.
4. Long Lead Material Procurement for Future System Deliveries
Procure graphics cards and FPGAs of the same batch order, to equip and prevent a design modification of systems to be delivered under CLINs 3.

CLIN 2 – Tier 2 Modeling Development

- 5. Systems Engineering and Program Management. IAW Appendix A:
 - a. Perform Delta SRR for Tier 2 Modeling Development and update RTVM for updated Tier 2 Modeling requirements considerations. Requirements shall be traceable to the Performance Specifications Document and the use case scenarios supporting overall system capability. (SRR A002, RTVM A008 CDRLs)
 - b. Perform Delta Design Review (DDR) to address critical design for Tier 2 Modeling Development for the CLIN 1 SG System and accompanying test plans for interim modeling capability achievements. (CDR A004, Test Plan & Procedures A014 CDRLs)
- 6. Tier 2 Model Application Development for baseline system and future growth
 - a. Design and develop Tier 2 Model Application for baseline system IAW EOIRDI Performance Specifications Document
 - b. Perform interim Tier 2 capability demonstration event.
 - c. Perform Tier 2 capability demonstration and FAT IAW Tier 2 requirements defined in the EOIRDI Performance Specification Doc and the RTVM, prior to integration with EOIRDI system. The FAT shall be performed using the Tier 2 model capability to demonstrate predefined EOIRDI SUT uses case scenarios with threat fly-out, and threat infrared signature models to confirm scene background, clutter, and atmospheric characteristics, as well as to confirm compatibility with the Baseline SKIP SGS modeling capabilities.
 - d. Perform Tier 2 model updates and integration for SG systems and integration as required during model application developmental cycle
 - e. Perform Tier 2 model demonstration during EOIRDI system FAT, SAT, IOC, or FOC.
- 7. SGS MANUFACTURER POST-DELIVERY SUPPORT. The Contractor shall provide post-delivery support for system integration /SAT, demonstration, training, system testing, system operation, system maintenance, ad hoc SG system updates, and the IOC/FOC site-testing for the Block A program phase.
 - a. ON-SITE. The contractor shall provide up to 40 hours total on-site support at NAS PAX River ACETEF
 - b. REMOTE. The contractor shall provide up to 50 hours of remote support

Option CLIN 3 – Air Force 6-Channel System

- 8. Systems Engineering and Program Management. IAW Appendix A:

- a. Perform Delta SRR and update RTVM to evaluate requirements potentially impacted by obsolescence issues associated with tech refresh/product improvement, and determine need for DDR. Requirements shall be traceable to the Performance Specifications Document and the use case scenarios supporting overall system capability. (SRR A002, RTVM A008 CDRLs)
 - b. If required due to obsolescence issues impacted by tech refresh/product improvement, perform ad hoc DDR of SG system design and accompanying test plans for interim technical confidence demonstrations. (CDR A004, Test Plan & Procedures A014 CDRLs)
9. Build of SG systems
 - a. Complete build of SG system to include a (1) Control and Modeling PC and 6 image generation channels
 - b. FAT of the scene generators and control and modeling PCs identified in item 10.a. The SG system Facility Acceptance Testing (FAT) procedure shall be performed on each completed asset before transfer to EOIRDI project sites. The purpose is to verify full functionality of SKIP using standard models and scenarios at 60Hz, 500Hz and 1000Hz, and frame size up to 2048x2048, in standalone and external control mode, with Use Case developed scenarios and signature validation, IAW SGS Performance requirements defined by the EOIRDI Performance Specifications Document
 - c. Transfer of 1 Control and Modeling PC and 6 Image Generation Channel to Lockheed Martin Corporation at Fort Worth for developmental work
10. SGS MANUFACTURER POST-DELIVERY SUPPORT.
 - a. The Contractor shall provide up to 80 hours of ad hoc phone support
 - b. The contractor shall provide up to 80 hours of on-site support for the Air Force to include support for software loads of completed (CLIN 1) Tier 1 and (CLIN 2) Tier 2 Modeling SW deliverables, system integration /SAT, demonstration, training, and FOC site-testing.

Option CLIN 4 – Terrain/Radiance Map Generation

11. RADIANCE MAP GENERATION. The contractor shall provide a system-integrated capability to generate 3D radiance maps for both fixed- and rotary-wing aircraft scenarios. These scenarios must include a wide variety of target area sizes (square miles) and clutter levels (polygons/second, statistical delta temperature) to simulate adverse battlefield conditions.
12. ACCEPTANCE TESTING AND DOCUMENTATION. The contractor must successfully demonstrate the EOIRDI system's ability to generate 3D radiance maps for both fixed-wing and rotary-wing aircraft scenarios. This demonstration must include a government-approved range of variable area sizes and clutter levels, as agreed upon between the government and contractor and documented in the CLIN 1 and 2 Test Plan and Procedure CDRL A014. Acceptance testing and documentation for this capability is included in CLINs 1 and 2 (FAT Report A015 CDRL, SAT Report A016 CDRL).
13. COORDINATION. The Contractor shall provide remote/virtual customer support and coordination for up to 20 meeting instances.

Option CLIN 5 – Additional User/Developer Documentation

14. EXPERT USER SYSTEM SUPPORT DOCUMENTATION. The contractor shall develop and deliver comprehensive user documentation and training materials for the EOIRDI system. These materials must be sufficient to train expert-level users and define processes/criteria for interfacing with the EOIRDI Scene Generation System for the long-term operational usability of the system. These documents shall supplement, and avoid duplication of content with, the CLINs 1 and 2 User's Manual (CDRL A017) and Operations and Maintenance Manual (CDRL A019).
- a. INTERFACE CONTROL DOCUMENT (ICD). The contractor shall develop and deliver an Interface Control Document (ICD). This document must detail the system's internal operability formats to allow for the integration of new models and system applications, enabling improved Scenario Generation System (SGS) operations and scenario development. (ICD CDRL A022)
 - b. FIRMWARE SUPPORT MANUAL (FSM). The contractor shall develop and deliver a Firmware Support Manual (FSM). The FSM must provide the necessary procedures to update the control FPGA firmware to support the future development and integration of potential new/updated SUT sensor emulator interfaces. (FSM CDRL A021)
15. COORDINATION. The Contractor shall provide remote/virtual customer support and coordination for up to 5 meeting instances.

Option CLIN 6 – ARINC Interface

16. The contractor shall provide an EOIRDI system-integrated ARINC-818 video output protocol for SUT-synchronized video transfer to supported threat processing units, as defined in the Performance Specifications Document.
17. ACCEPTANCE TESTING AND DOCUMENTATION. The contractor must successfully demonstrate the EOIRDI system's ability to transfer synchronized video to the SUT via the ARINC interface. This demonstration must meet the government-defined frame rates and sizes as detailed in the Performance Specification. Acceptance test planning, documentation, and testing for this capability is included in CLINs 1 and 2 (Test Plan and Procedure CDRL A014, FAT Report CDRL A015, SAT Report CDRL A016)
18. COORDINATION. The Contractor shall provide remote/virtual customer support and coordination for up to 5 meeting instances.

CLIN 7 – Spare Modeling PC - RESERVED**CLIN 8 – Spare Scene Generator - RESERVED****3.3 Additional Task Descriptions**

The following tasks are applicable across all of the above separate CLINs, and/or as specified uniquely within each individual task. All document deliverables shall be defined during CLINs 1 & 2, and reviewed and utilized as a baseline for updates if/as necessary, pertinent to CLIN 3, due to any external constraints/dependencies resulting in change to existing design or significant discrepancies in document content.

3.3.1 SYSTEMS ENGINEERING and PROGRAM MANAGEMENT (SEPM). The contractor shall provide the SEPM support to manage schedule and scope associated with the effort defined by this statement of work, Appendices, Performance Specifications Document, and accompanying CDRLs. The contractor shall provide data IAW Appendix A.

3.3.1.1 The contractor shall provide plans to execute the build of the baseline SGS and the modeling capability development, with demonstrations, as detailed within this SOW and accompanying SPEC. The contractor shall implement risk mitigation strategies and incremental technical progress/achievement in their technical approach, design, systems engineering approach and Life Cycle Sustainment planning as detailed within this SOW, accompanying Performance Specification, and Appendices. The contractor shall employ Reliability and Maintainability (R&M) engineering to influence system design in order to increase mission capability and availability and decrease logistics burden and cost over the system's life cycle by preventing or identifying R&M deficiencies early in development. The contractor shall deliver document and data deliverables IAW the following CDRLs: SEMP A011, R&M Plan A020, SDD A012, and Test Plan & Procedures A014.

3.3.1.2 The contractor shall coordinate, support, and document success criteria of each Milestone (MS) event IAW Appendix A (SOW/KO A001, SRR A002, PDR A003, CDR A004, TRR A005 CDRLs). Milestone reviews defined by Appendix A shall be conducted for CLIN 1, and observed at a Delta level for Tier 2/CLIN 2, and ad hoc for any external constraints that may result in change of design for CLIN 3.

3.3.1.3 The Contractor shall perform Risk Assessment and Management throughout the PoP of this effort. The Contractor shall use the Risk Assessment template, located in Appendix B, for briefings, MSRs, status updates, etc. The Contractor shall deliver document and data deliverables IAW MSR CDRL A007, Risk Management Plan (RMP) A013 CDRL, and Appendix A.

3.3.1.4 The contractor shall deliver the Scene Generation Systems (SGS) for operation within the broader EOIRDI capability under development; the SGS configuration and requirements are intended for use with multiple fixed and rotary winged Systems under Test (SUT) on Air Force, Army, and Naval airborne platforms.

3.3.1.5 The contractor shall provide and maintain a budget, expenditure plan, and an Integrated Master Schedule (IMS). (IMS CDRL A010, MSR CDRL A007)

3.3.1.6 The contractor shall track and provide updates on program status/performance, schedule/IMS, cost expenditures, and risk tracking in a monthly status report (MSR). (MSR CDRL A007)

3.3.2 TECHNICAL INTERCHANGE MEETINGS (TIMs)

3.3.2.1 The contractor shall support no fewer than two (2) technical interchange meetings (TIMs), one for CLIN 1/ Tier 1 and one for CLIN 2/ Tier 2, with the customer during interim development stages to provide updates on cost, schedule, capabilities, status, risk, and

preliminary system development test results. These TIMs shall be hosted either virtually or at the NGC Amherst site in Buffalo, NY.

- 3.3.2.2 The contractor shall provide minutes, agreed-upon action items, and interim progress reports as requested, from these meetings, and per each milestone/design review, to the government customer (Minutes CDRL A006).
- 3.3.3 COORDINATION. The Contractor shall support teleconferences, weekly for CLINs 1 & 2, and monthly for remaining CLINs, (either phone or video based) to discuss and maintain engineering and schedule progress, to assign actions to completing CDRL deliverables, and ensure the Government remains informed on program development.
- 3.3.4 RMF & REQTS TRACEABILITY. The Contractor shall support Risk Management Framework (RMF) and Requirements Traceability IAW DoDI 8510.01 RMF and associated guidelines to support successful deployment site Authorization to Operate (ATO), as agreed upon between the contractor and customer. Requirements shall be traceable to the Performance Specifications Document and the use case scenarios supporting overall system capability (RTVM A008, RMF A009 CDRLs).
- 3.3.5 EXECUTION & DELIVERY. The contractor shall support an incrementally funded project schedule with incremental build and delivery of SGSs, Tier 1 and 2 modeling development phases and corresponding milestones to assess schedule and technical progress and plan for final delivery as discussed in this SOW, Spec and CDRLs. An optional off-ramp for Tier 2 development shall be considered prior to Tier 2 execution. (SEMP A011, SDD A012 CDRLs)
- 3.3.6 EXTERNAL CONTROL TEST. The contractor shall support external SGS communication and data interface compliance testing of the SG systems. (Test Plan and Procedures A014 CDRL.)
- 3.3.7 USERS MANUAL: The contractor shall provide documentation to support customer site integration, maintenance and troubleshooting of the SG system (Users Manual A017 CDRL and O&M Manual A019 CDRL).
- 3.3.8 TEST & SITE INTEGRATION READINESS REVIEWS: The Contractor shall perform test and site integration readiness reviews and provide data and documentation. (TRR A005 CDRL)
- 3.3.9 SYSTEM TESTING AND EVALUATION: The Contractor shall execute SGS model demonstration testing and evaluation at Factory Acceptance Testing (FAT) and site acceptance testing (SAT) as approved by the Government in the RTVM and Test Plan, and deliver supporting data and documentation IAW the criteria defined in Appendix A. (Test Plan and Procedures A014 CDRL, FAT Report A015 CDRL, SAT Report A016 CDRL)
- 3.3.9.1 Factory Acceptance Test (FAT)
The Contractor shall conduct a FAT at the contractor's facility following fabrication and sub-system factory testing to review test results to verify that the system deliverables are ready for delivery to the government for installation and final acceptance testing. The contractor shall present draft FAT procedures at the CDR and the final version at TRR (Test Plan & Procedures CDRL A014). The FAT shall emulate, as much as possible, System Under Test (SUT) conditions and connections, the government use cases, and the SAT test environment to validate the system functionality, and to discover problems needing corrective action prior to the Site Acceptance Test for smooth transition from FAT to SAT.

The Government will witness contractor testing of the deliverable at the Contractor facility prior to shipping. The Contractor shall provide a test report at the conclusion of testing (FAT Report A015 CDRL). The Government will conduct final system acceptance testing at a government site, respective to each CLIN. Acceptance testing will determine if the deliverables meet the performance requirements. Some acceptance testing activities may use results of prior testing. The Contractor shall support acceptance testing by providing personnel to operate the system and provide hands on training in the operation and maintenance of the system, as defined per each CLIN.

3.3.9.2 Site Acceptance Testing (SAT)

The system shall be delivered to the Government site, respective to each CLIN. The SAT for each test capability will be conducted using operational test equipment and/or platforms as available at the Government site. The Contractor shall provide draft SAT procedures at the CDR and the final version at TRR (Test Plan & Procedures CDRL A014). The Government will coordinate Government site test assets and general test equipment. The SAT will have two parts. The first part is a re-execution of portions of the FAT after shipping and assembly, with tests that re-demonstrate key parts of the FAT related to operations, as mutually agreed upon at TRR. The second part of the SAT will include everything required to verify the system performance in accordance with the system RTVM and government use cases, for each test capability. The contractor shall report the results of the SAT after the conclusion of testing (SAT Report A016 CDRL)

3.3.10 TECHNICAL DATA: The Contractor shall deliver data and documentation in support of a Technical Data Package (TDP) per site deliverable (SDD, Products, Drawings, Associated Lists [PDL A018 CDRL])

3.4 Equipment for SG System testing

1. Inter-Range Instrument Control (IRIG) for facility time (facility supplied)
2. Government Furnished Next Generation Threat System for scenario development
3. Government Furnished Stimulation System Integration System (StimSYS) for scenario external control of scene generation system and System-Under-Test (SUT)
4. Master Control and Modeling Scene Generator, manufacturer supplied
5. SG Image Rendering Channels, manufacturer supplied
6. Government Furnished Motion Simulation Console for SUT control
7. Government Furnished Models for SG Scenario Development
 - a. Terrain using the SWUSA database
 - b. Atmospheric database with 11 available standard MODTRAN models
 - c. Classified TETRA T&E Threat models at integration sites

N0042122G0004

N0042123F0031

Page 12 of 20

Appendix A – Milestone Review Entry and Exit Criteria and Accompanying Data Products

The following section describes all reviews with respective entry and exit criteria for the incremental build of the SKIP baseline SGS in coordination with external EOIRDI system development and for the tiered modeling capability updates as discussed in this SOW. The purpose of these reviews is to track progress and ensure the system performance requirements are met, as captured in the system performance specification and the overall use cases of the overall EOIRDI project. Additional guidance for Risk Assessment and Management Planning is located in Appendix B.

D.1 Start of Work / Kick Off (SOW/KO)

The SOW/KO is a conversation between the respective Government and Contractor teams intended to confirm that the project is properly planned and ready for execution, and to present an overview of the Existing Baseline SKIP capability, status, and product specifications to identify project baseline for Government approval and procurement initiation, as/if applicable. A SOW/KO is required at the start of the contract and a delta SOW/KO is required at the start of each successive CLIN. Deliverables at this stage are at a preliminary level for conceptual planning purposes. SOW/KO may be combined with SRR.

a. Entry Criteria

- i. A preliminary agenda has been coordinated (nominally) 15 days prior to the SOW/KO.
- ii. Supplementation materials have been made available to participants prior to the SOW/KO:
 - (1) Organization Chart with POCs and contact info
 - (2) Contract Work Breakdown Structure (CWBS) and CWBS dictionary
 - (3) IMS
 - a. include proposed Build of Systems planning to meet emulator DT schedule
 - b. Modeling S/W development schedule including a tiered development approach with scheduled demonstrations
 - (4) Budget and Spend Plan
 - (5) Comprehensive risk assessment
 - (6) Detailed overview of Baseline/Existing SKIP Capability (Tier 0) with building infrastructure coordination, as applicable
 - (7) Baseline/Existing SKIP SGS product specifications for each hardware and software configuration item and procurement details
 - (8) Initial RTVM to trace Baseline/Existing system requirements / RFI evaluation discussion. Detail what requirements have been met for:
 - a. Baseline/Existing (Tier 0)
 - b. Tier 1
 - c. Tier 2

- (9) Technological Assurance of Baseline/Existing KEOS/SKIP Capability to demonstrate H/W capability and Modeling building requirements as a functional integrated component of the EOIRDI system

b. SOW/KO Exit Criteria

- i. Completion of the SOW/KO should include the following:
 - (1) Result of the Baseline/Existing SKIP SGS/Modeling Capability Technological Assurance provides high confidence of performance specification requirements for Tier 0 and functional integration with the EOIRDI system
 - (2) Baseline/Existing system requirements satisfaction (requirements traceability) are tracked in the initial RTVM
 - (3) The initial program baseline, WBS, budget, schedule, and spend plan are executable and the SGS build schedule accommodates known emulator DT schedule
 - (4) GOVT concurrence of IMS, CWBS, and Spend Plan
 - (5) Concurrence of Action Items and Issues identified during SOW/KO

c. SOW/KO Products

- i. SOW/KO documentation to include Baseline/Existing SKIP SGS design, technological assurance of interoperability with EOIRDI system, and building infrastructure considerations
- ii. Baseline/Existing SKIP SGS product specifications for each hardware and software configuration item and procurement plan
- iii. Each SOW/KO participant provided with the following documents:
 - (1) List of attendees, to include name, functional area represented, phone number, and email address
 - (2) Meeting minutes
 - (3) Recommendation as to the technical readiness of the program to enter the next phase of development
- iv. Updated IMS, CWBS, and spend plan
- v. Existing Baseline SKIP system external IDD
- vi. Existing Baseline SKIP system IDD/ICDs, Users Manuals v.1

D.2 SRR

The purpose of the SRR is to assess and ensure the existing SKIP SGS solution with tiered model building development is consistent with the system solution, as identified in the performance specification document.

a. SRR Entry Criteria

- i. A preliminary agenda has been coordinated (nominally) 15 days prior to the SRR.
- ii. SRR technical products for hardware and software configuration item have been made available to the cognizant SRR participants prior to the review:
 - (1) All requirements have been incorporated in the RTVM

- (2) Refined System Performance Specification
- (3) Proposed solution to satisfy requirements and capability iterated in the System Performance Specification
- (4) Address RMF considerations
- (5) Proposed analysis of reliability, maintainability, and availability (RM&A) of system (Life Cycle discussion)
- (6) Discuss considerations for system operation and storage at various classification levels
- (7) Proposed system design Human Systems Integration (HSI) considerations for both SW and HW
- (8) An updated comprehensive risk assessment and management plan
- (9) Updated program schedule and spend plan

b. SRR Exit Criteria

- i. Mutually understood system requirements, specifications and technical approach in support of system functional definition, functional decomposition, test, and evaluation
- ii. Executable program schedule and budget, considering risks and constraints
- iii. Mutually understood reliability, maintainability, and availability (RM&A) considerations
- iv. Comprehensive risk assessment and management plan

c. SRR Products

- i. SRR documentation materials
- ii. Completion of the SRR should provide the following:
 - (1) An approved System Performance Specification
 - (2) An approval of all system performance requirements identified in the RTVM
 - (3) Overviews of proposed solution and technical approach
 - (4) An updated and approved program schedule and budget that addresses critical path drivers
 - (5) An updated and approved program and technical staffing resource list (organization chart)
 - (6) Approved RM&A evaluation
 - (7) An updated and approved and comprehensive risk assessment and management plan, including risks and mitigation approach options
- iii. Each SRR participant provided with the following documents:
 - (1) List of attendees, to include name, functional area represented, phone number, and email address
 - (2) Meeting minutes
 - (3) Mutually agreed upon action items (RFAs) with updated statuses
 - (4) Recommendation as to the technical readiness of the program to enter the next phase of development.

D.3 PDR

The purpose of the PDR is to evaluate detailed design, performance, cost, schedule, and risk requirements of the existing SKIP SGS solution with tiered model building development.

a. PDR Entry Criteria

- i. The program SRR has been successfully completed and SRR exit criteria and RFAs have been satisfied
- ii. A preliminary agenda has been coordinated (nominally) 15 days prior to the PDR
- iii. PDR technical products of the model building software configuration have been made available to the cognizant PDR participants prior to the review:
 - (1) Updated RTVM with proposed verification methods
 - (2) Presentation of existing SKIP SGS HW configuration and Baseline/Tier 0 design
 - (3) Preliminary design of both Tier 1 and Tier 2 modeling capabilities' implementation to include internal specifications, conceptual configurations, data/IO flow charts, and HSI
 - (4) Preliminary model and scenario building methodology supports functional implementation of performance specifications; design of incremental feature updates does not adversely affect baseline system performance
 - (5) Preliminary Tier 1 and Tier 2 developmental approach and SW configuration management processes and plan
 - (6) Preliminary Incremental Capability Demonstrations and Training plan
 - (7) Potential impact of SW design on reliability, maintainability, and availability (RM&A) of system has been considered at a preliminary level
 - (8) Preliminary documentation related to the RMF process
 - (9) Updated risk assessment and management plan
 - (10) Critical path analysis based on updated IMS

b. PDR Exit Criteria

- i. The preliminary design, as disclosed, satisfies the system performance specification and use cases
- ii. The status of the technical effort, design, development approach, and methodology indicates operational evaluation success (operationally suitable and effective); design of incremental feature updates does not adversely affect baseline system performance
- iii. The preliminary SW configuration processes are viable and ensure successful development and sustainment of coding best practices
- iv. The program baseline is functional and the program spend plan and schedule is executable and considers risks and constraints.
- v. Incremental Capability Demonstrations and Training plan has been defined at the preliminary level and incorporated into the IMS
- vi. Completion of critical path analysis based on updated IMS

- vii. Potential impact of SW design on reliability, maintainability, and availability (RM&A) of system has been identified and addressed at a preliminary level
- viii. RMF process is being properly tracked within the schedule and all upcoming activities are being scheduled by all project teams involved
- ix. All cost, schedule, technical, and performance risks have been mitigated to a “low” or “moderate” risk classification status (probability/likelihood vs impact/consequence)
- x. The contractor has submitted drawings and other support documentation that illustrates the design and any facility infrastructure requirements

c. PDR Products

- i. PDR documentation materials
- ii. Each PDR participant is provided with the following documents:
 - (1) List of attendees, to include name, functional area represented, phone number, and email address
 - (2) Mutually agreed upon action items (RFAs) with updated statuses
 - (3) Meeting minutes
- iii. An updated and approved IMS, including critical path analysis
- iv. An approved preliminary design, technical and development approach, implementation and methodology
- v. An approved preliminary SW configuration management approach and processes
- vi. An approval of all system performance requirements identified in the RTVM
- vii. An approved preliminary RM&A plan
- viii. An updated and approved and comprehensive risk assessment and management plan, including risks and mitigation approach options
- ix. Recommendation as to the technical readiness of the program to enter the next phase of development

D.4 CDR

The purpose of the CDR is to evaluate detailed design, performance, cost, schedule, and risk requirements of the existing SKIP SGS solution with tiered model building development to ensure system is ready to proceed with Tiers 1 and 2 of modeling capability development, demonstration, and subsystem and system test.

a. CDR Entry Criteria

- i. The program PDR has been successfully completed and PDR exit criteria and RFAs have been satisfied
- ii. A preliminary agenda has been coordinated (nominally) 15 days prior to the CDR.
- iii. CDR technical products (hardware and software elements of the product baseline to be reviewed and approved at the CDR) have been made available to the cognizant CDR participants prior to the review
- iv. Final design includes existing SKIP SGS HW configuration and Baseline/Tier 0 capability
- v. Final design of both Tier 1 and Tier 2 modeling capabilities' implementation to include internal specifications, configurations, data/IO flow charts, and HSI

- vi. Final model and scenario building methodology demonstrates functional implementation of performance specifications; design of incremental feature updates does not adversely affect baseline system performance
 - vii. Finalized Tier 1 and Tier 2 developmental approach and SW configuration management processes and plan
 - viii. System Performance Requirements tracking in an updated RTVM with traceability of proposed verification methods
 - ix. Updated product specifications for each hardware and software configuration item
 - x. Final System modeling capability specification
 - xi. Preliminary test procedures for finalized Incremental Capability Demonstrations and Training plan
 - xii. Updated risk assessment and management plan to include impact of designed Tier 1 and Tier 2 SW upgrades on reliability, maintainability, and availability (RM&A) of system
 - xiii. Review SEMP changes, if any
 - xiv. Updated documentation related to the RMF process.
- b. CDR Exit Criteria
- i. The status of the technical effort, design, and development approach indicates high likelihood of operational success (operationally suitable and effective)
 - ii. The detailed design (hardware and software), as disclosed, satisfies the system performance specification and use case implementation; design of incremental feature updates does not adversely affect baseline system performance
 - iii. The final SW configuration processes are viable and ensure successful development and sustainment of coding best practices
 - iv. The program budget, spend plan and schedule is functional and executable within the approved product baseline
 - v. The program is properly staffed
 - vi. Completion of critical path analysis based on updated IMS
 - vii. An acceptable level of program risk is confirmed
 - viii. Incremental Capability Demonstrations with Training plan are defined and incorporated into the IMS. Corresponding test procedures meet program scope for incremental technical progress requirements
 - ix. RMF documentation and scheduling is ready to support (Assessment & Authorization) A&A (ATO preparation), as applicable
 - x. The contractor has submitted drawings and other support documentation that illustrates the design and any facility infrastructure requirements
 - xi. All system requirements and specifications are approved and traceable within the RTVM
 - xii. The risks are known and manageable
- c. CDR Products
- i. CDR documentation materials
 - ii. Each CDR participant is provided with the following documents:
 - (1) List of attendees, to include name, functional area represented, phone number, and email address

- (2) Mutually agreed upon action items (RFAs) with updated statuses
- (3) Meeting minutes
- (4) Recommendation as to the technical readiness of the program to enter the next phase of development
- (5) An established system product baseline
- iii. An approval of all system performance requirements identified in the RTVM
- iv. An approved and comprehensive risk assessment and management plan with agreed upon risks and mitigation approach
- v. An approved critical design, technical approach, and SW implementation and methodology
- vi. An approved and updated program development schedule, software development plan, T&E with demonstration events, scheduled system software updates, and critical path drivers
- vii. An approved SW configuration management approach and processes
- viii. Detailed software configuration, software process flowchart, functional and/or timing diagrams, and drawings package
- ix. Updated IMS showing proposed fabrication, testing events, reviews, site delivery, site integration, Government acceptance testing, and critical path analysis.
- x. Recommendation as to the technical readiness of the program to enter the next phase of development

D.5 TRR

The purpose of the TRR is to ensure the system is ready to begin formal acceptance testing by the Government. The TRR will ensure readiness of the SGS and developed Tier 1 and Tier 2 Modeling and Scenario Development SW to be integrated as a functional part of the EOIRDI system. The TRR will also ensure readiness for integration with the applicable host site facility in support of required Contractor and Integrated test events. A TRR for each Tier 1 and 2 deliverables is required to be demonstrated on the host SGS.

a. TRR Entry Criteria

- i. Configuration and definition of FAT activities to be performed
- ii. All applicable and functional, unit level, subsystem, system, and integration tests were conducted successfully before FAT
 - a. Test constraints based on previous testing or test hardware limitations have been identified
 - b. Test configuration (test article and instrumentation and support equipment) has been identified
 - c. Detailed test procedures have been established
 - d. Plans for collection, reduction, and analysis of the test data have been established.
 - e. System calibration plans and calibration status confirmation have been established.
- iii. All TRR specific materials, such as test plans, test cases, and procedures, are available to all participants prior to conducting the review (minimum of seven working days prior to commencement of testing)

- iv. Test procedures comply with plans; test procedure descriptions are sufficient to perform test execution tasks and to ensure verification methodology will satisfy system requirements
 - v. FAT procedures are documented and confirmed to be sufficient to satisfy all requirements
 - vi. RTVM documentation is current for requirement verification and traceability
 - vii. A list of currently open problem reports, both external and internal, cross-referenced to the contracted deliverable end items or development hardware and software products has been generated
 - viii. Documented model software design changes has been generated
 - ix. For RMF requirements, pre-assessments scans (within 30 days minimum) have been completed; finalized RMF-related CDRL documentation (within 15 days minimum) has been submitted
- b. TRR Exit Criteria
- i. RTVM review confirms that all requirements have traceability to test procedures
 - ii. Facility infrastructure support considerations have been provided
 - iii. Completed and approved FAT plans and procedures have been provided
 - iv. Completed and approved test plans and procedures for integrated EOIRDI System Testing
 - v. Identification and coordination of required test resources is completed
 - vi. Completed and approved SAT plans and procedures have been provided
- c. TRR Products
- i. Approval to proceed to FAT
 - ii. Completed and approved Updated User and Operations Manual
 - iii. Completed and approved Contractor FAT and SAT test plans and procedures